

Flow charts and operations



1 $s \xrightarrow{-7} s - 7 \xrightarrow{\times 6} 6(s - 7)$

2 $t \xrightarrow{+3} t + 3 \xrightarrow{\div 5} \frac{(t + 3)}{5}$

3 a $y \xrightarrow{\times 5} 5y \xrightarrow{+8} 5y + 8$

b $k \xrightarrow{\div 3} \frac{k}{3} \xrightarrow{-7} \frac{k}{3} - 7$

c $p \xrightarrow{+15} p + 15 \xrightarrow{\times 8} 8(p + 15)$

d $n \xrightarrow{+5} n + 5 \xrightarrow{\div 3} \frac{n + 5}{3}$

e $x \xrightarrow{\times 4} 4x \xrightarrow{+3} 4x + 3$

f $j \xrightarrow{\div 9} \frac{j}{9} \xrightarrow{-7} \frac{j}{9} - 7$

g $p \xrightarrow{+28} p + 28 \xrightarrow{\times 3} 3(p + 28)$

h $s \xrightarrow{-6} s + 6 \xrightarrow{\times 7} 7(s + 6)$

Operations in reverse order

4 a $6y + 9 \xrightarrow{-9} 6y \xrightarrow{\div 6} y$

b $5(q + 3) \xrightarrow{\div 5} q + 3 \xrightarrow{-3} q$

c $6(s - 9) \xrightarrow{\div 6} s - 9 \xrightarrow{+9} s$

d $\frac{n + 8}{5} \xrightarrow{\times 5} n + 8 \xrightarrow{-8} n$

e $\frac{j}{5} - 6 \xrightarrow{+6} \frac{j}{5} \xrightarrow{\times 5} j$

f $8y + 9 \xrightarrow{-9} 8y \xrightarrow{\div 8} y$

5

a $4(q+9) \xrightarrow{\div 4} q+9 \xrightarrow{-9} q$



b $5(r-7) \xrightarrow{\div 5} r-7 \xrightarrow{+7} r$

c $\frac{n+6}{9} \xrightarrow{\times 9} n+6 \xrightarrow{-6} n$

d $\frac{j}{2} - 7 \xrightarrow{+7} \frac{j}{2} \xrightarrow{\times 2} j$

Two-step equations

6

a Operation 1: Subtract 16
Operation 2: Divide by 4

b $24 \xrightarrow{\text{Operation 1}} 8 \xrightarrow{\text{Operation 2}} 2$

c $x = 2$

7

a Operation 1: Divide by 3
Operation 2: Subtract 11

b $57 \xrightarrow{\text{Operation 1}} 19 \xrightarrow{\text{Operation 2}} 8$

c $p = 8$

8

a Operation 1: Add 11
Operation 2: Multiply by 7

b $-7 \xrightarrow{\text{Operation 1}} 4 \xrightarrow{\text{Operation 2}} 28$

c $j = 28$

9

a i Subtract 6 from both sides, then divide both sides by 7.

ii $p = 5$

b i Add 5 to both sides, then divide both sides by 2.

ii $x = 8$

c i Divide both sides by 5, then add 15 to both sides.

ii $n = 22$

d i Divide both sides by 10, then subtract 10 from both sides.

ii $p = 2$



e

i Subtract 3 from both sides, then multiply both sides by 8.

ii $y = 16$

f

i Add 11 to both sides, then multiply both sides by 3.

ii $x = 60$

g

i Multiply both sides by 6, then subtract 22 from both sides.

ii $u = 2$

h

i Multiply both sides by 2, then add 10 to both sides.

ii $y = 20$